localCloudConfiguration HTTP/TLS/JSON

Interface Design Description

Service ID: *"localCloudConfiguration"*

Abstract
This document describes the HTTP/TLS/JSON interface of the `localCloudConfiguration` service, version 1.0.0, of the LocalCloudConfigurator support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `localCloudConfiguration` service interface of the LocalCloudConfigurator system.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `localCloudConfiguration` service interface

This document provides the Interface Design Description IDD to the *localCloudConfiguration – Service Description* document [5].

2.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, port 8492, base path `/api` and the service metadata `systemVersion=5.2.2`. Path parameters are templates (`{id}`):

Operation	Method	Path
`catalog`	GET	`/{}catalog`
`status`	GET	`/{}status`
`plan`	POST	`/{}plan`
`submit-deployment`	POST	`/{}deployments`
`get-deployment`	GET	`/{}deployments/{}{id}`
`list-configurations`	GET	`/{}configurations`
`create-configuration`	POST	`/{}configurations`
`get-configuration`	GET	`/{}configurations/{}{id}`
`update-configuration`	PUT	`/{}configurations/{}{id}`
`remove-configuration`	DELETE	`/{}configurations/{}{id}`

3 Service Operations

Each subsection names the HTTP method and path of an operation, the abstract operation of the SD and its input and output types.

Every request needs the caller's client certificate (mutual TLS) and `Authorization: Bearer IDENTITY-TOKEN//<token>` of the same system; the operation is allowed for Sysop, the systems of `lcc.api.whitelist` and systems with a ConsumerAuthorization grant of the operation. These checks answer 401 and 403 with the common error body (`errorMessage`, `errorCode`, `exceptionType`, `origin`); other `/api` paths are refused (403) before any handler. Request bodies are JSON of at most 64 KiB (413 `"Request exceeds 64 KiB"`); unknown fields are rejected (400). Errors of the operations are `{"error": "<text>"}` with 400 (invalid input), 404, 409 (stale plan, running deployment, changed design), 413 and 502 (backend failure, redacted message). Responses carry `Cache-Control: no-store`, `X-Content-Type-Options: nosniff` and `Content-Security-Policy: default-src 'none'; frame-ancestors 'none'`. Absent `null` fields are omitted.

3.1 GET /{}api/{}catalog

Operation: catalog
Output: Catalog

The component catalog of the cloud.

A successful call returns `200 OK`.

```
GET /api/catalog HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
```

Listing 1: An catalog invocation.

```
{
  "cloud": "TestCloud",
  "composeProject": "arrowhead-v52-test-cloud",
  "colony": "arrowhead-java-localcloud",
  "executorType": "arrowhead-java-cloud-executor",
  "defaults": ["serviceregistry", "authentication",
               "consumerauthorization", "serviceorchestration-dynamic"],
  "smoke": ["verify-localcloud"],
  "systems": [{"id": "timemanager", "name": "TimeManager", "kind": "support",
               "services": ["timemanager-db", "timemanager"],
               "internalAssets": ["timemanager-db"],
               "requires": ["amqp-broker", "consumerauthorization"],
               "provision": ["provision-timemanager-identity.sh"],
               "policies": ["configure-timemanager-policy.sh"],
               "protected": false}, ...]
}
```

Listing 2: An catalog response.

3.2 GET /{}api/{}status

Operation: status
Output: Status

Running Compose services and the state (`running`, `partial`, `stopped`) of each component (cached 2 s).

A successful call returns `200 OK`. Errors: 502 when the backend cannot report the runtime.

```
GET /api/status HTTP/1.1
Host: localcloudconfigurator:8492
```

```
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
```

Listing 3: An status invocation.

```
{
  "cloud": "TestCloud",
  "running": ["serviceregistry-db", "serviceregistry", ...],
  "systems": [{"id": "serviceregistry", "state": "running"},
              {"id": "timemanager", "state": "stopped"}, ...]
}
```

Listing 4: An status response.

3.3 POST /{}api/{}plan

Operation: plan
Input: PlanRequest
Output: PlanResponse

Builds a plan against the current runtime; returns it with its hash.

A successful call returns `200 OK`. Errors: 400 for an unknown component, a forbidden removal (protected, selected, still required) or a dependency cycle.

```
POST /api/plan HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
Content-Type: application/json

{
  "systems": ["timemanager"],
  "remove": [],
  "reapply": false,
  "smoke": true
}
```

Listing 5: An plan invocation.

```
{
  "plan": {
    "cloud": "TestCloud",
    "catalogHash": "9f2c...",
    "systems": ["amqp-broker", "serviceregistry", ..., "timemanager"],
    "steps": [{"kind": "provision", "target": "provision-timemanager-identity.sh",
               "system": "timemanager"},
              {"kind": "up", "target": "timemanager-db", "system": "timemanager"},
              {"kind": "up", "target": "timemanager", "system": "timemanager"},
              {"kind": "policy", "target": "configure-timemanager-policy.sh",
               "system": "timemanager"},
              {"kind": "smoke", "target": "verify-localcloud"}],
    "notes": ["ServiceRegistry: already running", ...]
  },
  "hash": "41d8..."
}
```

Listing 6: An plan response.

3.4 POST /{}api/{}deployments

Operation: submit-deployment
Input: ApplyRequest
Output: DeploymentAccepted

Applies a reviewed plan if its hash still matches a fresh plan and no other deployment is running.

A successful call returns `202 Accepted` (`200 OK` when the plan has no steps). Errors: 409 `"Plan or cloud state changed. Review a fresh plan before applying."` or `"A deployment is still running. Wait for it before applying another plan."`; 400 as for plan; 502 when the backend fails.

```
POST /api/deployments HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
Content-Type: application/json

{
  "systems": ["timemanager"],
  "remove": [],
  "reapply": false,
  "smoke": true,
  "expectedPlanHash": "41d8..."
}
```

Listing 7: An submit-deployment invocation.

```
{
  "id": "local-0f8e2a1c-6d4b-4a3e-9c1d-2b7f5e8a9c01",
  "plan": { ... },
  "state": "submitted"
}
```

Listing 8: An submit-deployment response.

3.5 GET /{}api/{}deployments/{}*{id}*

Operation: get-deployment
Input: Identifier
Output: Deployment

State and step results of a submitted deployment (identifier appended to the path); the report is stored.

A successful call returns `200 OK`. Errors: 404 `"Deployment not found"` for an identifier this system did not submit (or a historical ColonyOS workflow with the direct backend).

```
GET /api/deployments/local-0f8e2a1c-6d4b-4a3e-9c1d-2b7f5e8a9c01 HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
```

Listing 9: An get-deployment invocation.

```
{
  "id": "local-0f8e2a1c-6d4b-4a3e-9c1d-2b7f5e8a9c01",
  "state": "running",
  "steps": [{"node": "s1", "processId": "", "kind": "up",
             "target": "timemanager-db", "state": "success",
             "errors": [], "output": "..."},
            {"node": "s2", "processId": "", "kind": "up",
             "target": "timemanager", "state": "running",
             "errors": [], "output": ""}]
```

```
}
```

Listing 10: An get-deployment response.

3.6 GET /{}api/{}configurations

Operation: list-configurations
Output: ConfigurationList

Saved designs (up to 1000), sorted by name.
A successful call returns `200 OK`.

```
GET /api/configurations HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
```

Listing 11: An list-configurations invocation.

```
{
  "entries": [{"id": "3f1c9a7e-2b4d-4e6f-8a1b-9c0d2e3f4a5b", "name": "Line 1", ...}]
}
```

Listing 12: An list-configurations response.

3.7 POST /{}api/{}configurations

Operation: create-configuration
Input: ConfigurationDraft
Output: Configuration

Saves a new design (revision 1).
A successful call returns `201 Created`. Errors: 400 for an invalid name, components or positions.

```
POST /api/configurations HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
Content-Type: application/json

{
  "name": "Line 1",
  "systems": ["timemanager"],
  "remove": [],
  "layout": {"timemanager": {"x": 320, "y": 140}}
}
```

Listing 13: An create-configuration invocation.

```
{
  "id": "3f1c9a7e-2b4d-4e6f-8a1b-9c0d2e3f4a5b",
  "name": "Line 1",
  "systems": ["serviceregistry", "authentication", "consumerauthorization",
              "serviceorchestration-dynamic", "timemanager"],
  "remove": [],
  "reapply": false,
  "smoke": false,
  "layout": {"timemanager": {"x": 320.0, "y": 140.0}},
  "revision": 1,
  "updatedAt": "2026-11-09T10:00:00.123456Z"
}
```

Listing 14: An create-configuration response.

3.8 GET /{}api/{}configurations/{}*{id}*

Operation: get-configuration
Input: Identifier
Output: Configuration

One saved design.

A successful call returns `200 OK`. Errors: 404 `"Configuration not found"`.

```
GET /api/configurations/3f1c9a7e-2b4d-4e6f-8a1b-9c0d2e3f4a5b HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
```

Listing 15: An get-configuration invocation.

3.9 PUT /{}api/{}configurations/{}*{id}*

Operation: update-configuration
Input: ConfigurationDraft
Output: Configuration

Replaces a design when `expectedRevision` matches; the revision is incremented.

A successful call returns `200 OK`. Errors: 409 `"Configuration changed; reload before saving"`; 404; 400.

```
PUT /api/configurations/3f1c9a7e-2b4d-4e6f-8a1b-9c0d2e3f4a5b HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
Content-Type: application/json

{
  "name": "Line 1 (edited)",
  "systems": ["timemanager"],
  "expectedRevision": 1
}
```

Listing 16: An update-configuration invocation.

3.10 DELETE /{}api/{}configurations/{}*{id}*

Operation: remove-configuration
Input: RevisionCheck
Output: Removal

Deletes a design when `expectedRevision` matches; the runtime is unchanged.

A successful call returns `200 OK`. Errors: 409 `"Configuration changed; reload before deleting"`; 404.

```
DELETE /api/configurations/3f1c9a7e-2b4d-4e6f-8a1b-9c0d2e3f4a5b HTTP/1.1
Host: localcloudconfigurator:8492
Authorization: Bearer IDENTITY-TOKEN//<token of LocalCloudManager>
Content-Type: application/json

{
```

```
  "expectedRevision": 2
}
```

Listing 17: An remove-configuration invocation.

```
{
  "removed": "3f1c9a7e-2b4d-4e6f-8a1b-9c0d2e3f4a5b"
}
```

Listing 18: An remove-configuration response.

4 Data Models

The JSON objects of the interface, in alphabetic order.

4.1 struct ApplyRequest

Field	Type	Description
systems	List<Name>	Selected components.
remove	List<Name>	Components to stop.
reapply	Boolean	Rerun setup of running components.
smoke	Boolean	Add the verification step.
expectedPlanHash	String	Hash of the reviewed plan (409 if it no longer matches).

4.2 struct Catalog

Field	Type	Description
cloud	Name	Local cloud.
composeProject	Name	Compose project.
colony	Name	ColonyOS colony (legacy backend).
executorType	Name	ColonyOS executor type.
defaults	List<Name>	Default core components.
smoke	List<Name>	Verification operations (`verify-localcloud`).
systems	List<Component>	Components.

4.3 struct Component

Field	Type	Description
id	Name	Component identifier.
name	String	Display name.
kind	String	`core`, `support`, `application` or `infrastructure`.
services	List<Name>	Compose services, databases first.
internalAssets	List<Name>	Owned databases.
requires	List<Name>	Dependencies.
provision	List<String>	Provisioning scripts.
policies	List<String>	Policy scripts.
protected	Boolean	Cannot be stopped.

4.4 struct Configuration

Field	Type	Description
id	Identifier	UUID.
name	String	1–200 characters.
systems	List<Name>	Defaults first, then the selection (databases mapped to their host).
remove	List<Name>	Components marked for stopping.
reapply	Boolean	
smoke	Boolean	
layout	Object<Point>	Positions by component.
revision	Number	Incremented on every update.
updatedAt	DateTime	Time of the last change.

4.5 struct ConfigurationDraft

Field	Type	Description
name	String	1–200 characters.
systems	List<Name>	
remove	List<Name>	
reapply	Boolean	
smoke	Boolean	
layout	Object<Point>	At most 200 positions.
expectedRevision	Number	Required for update-configuration.

4.6 struct ConfigurationList

Field	Type	Description
entries	List<Configuration>	Saved designs.

4.7 struct Deployment

Field	Type	Description
id	Identifier	Deployment identifier.
state	String	`running`, `success` or `failed` (ColonyOS: also `waiting`).
steps	List<StepState>	One entry per plan step.

4.8 struct DeploymentAccepted

Field	Type	Description
id	Identifier	Deployment identifier (empty when there was nothing to do).
plan	Plan	The applied plan.
state	String	`submitted`, or `success` when the plan had no steps.

4.9 struct Plan

Field	Type	Description
cloud	Name	Local cloud.
catalogHash	String	SHA-256 of the catalog.
systems	List<Name>	Components in dependency order.
steps	List<Step>	Ordered steps.
notes	List<String>	For example `"TimeManager: already running"`.

4.10 struct PlanRequest

Field	Type	Description
systems	List<Name>	Selected components (at most 200).
remove	List<Name>	Components to stop (at most 200).
reapply	Boolean	
smoke	Boolean	

4.11 struct PlanResponse

Field	Type	Description
plan	Plan	The plan.
hash	String	SHA-256 of the plan.

4.12 struct Point

Field	Type	Description
x	Number	0–10000.
y	Number	0–10000.

4.13 struct Removal

Field	Type	Description
removed	Identifier	Removed design.

4.14 struct RevisionCheck

Field	Type	Description
expectedRevision	Number	Current revision.

4.15 struct Status

Field	Type	Description
cloud	Name	Local cloud.
running	List<Name>	Running Compose services.
systems	List<Object>	`id` and `state` per component.

4.16 struct Step

Field	Type	Description
kind	String	`provision`, `up`, `policy`, `stop` or `smoke`.
target	String	Script, Compose service or verification.
system	Name	Owning component (absent for `smoke`).

4.17 struct StepState

Field	Type	Description
node	String	Step identifier (`s1`, ...).
processId	String	ColonyOS process (empty for the worker).
kind	String	Step kind.
target	String	Step target.
state	String	`waiting`, `running`, `success` or `failed`.
errors	List<String>	Redacted errors.
output	String	Redacted, bounded command output.

4.18 Primitives

Type	Description
Boolean	JSON Boolean.
DateTime	JSON String, ISO 8601 UTC timestamp (`Instant.toString`).
Identifier	JSON String.
List	JSON Array [4].
Name	JSON String.
Number	JSON Number.
Object	JSON Object.
String	JSON String.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-LocalCloudConfigurator-go`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. This is a native operator API/UI using `net/http` and ColonyOS. The `/api` routes require an operator bearer token, not an Arrowhead identity token. It is a distinct binding from the Java support-system wrapper. `Eclipse-Arrowhead-go/ah5-LocalCloudConfigurator-go/internal/api/api.go` records the handlers.

Listening port. serve listens on 127.0.0.1:8095 by default. Override with -listen host:port, SERVER_PORT=N or -server.port N; the port overrides keep the -listen host. For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

No Rust implementation folder for LocalCloudConfigurator is present in the designated primary Rust tree. There is no port-specific executable, service binding or parity claim in this edition. A future port needs its own build, state model, security and interface qualification.

6 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “localCloudConfiguration – Service Description (LocalCloudConfigurator),” Eclipse Arrowhead, SD 5.3.0, 2026, localCloudConfigurator-v5.3.0-localCloudConfiguration-SD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-11-09	5.2.2	Document created for the `localCloudConfiguration` HTTP/TLS/JSON interface of the Java-Spring LocalCloudConfigurator (port 8492).	Claude Opus 5.5 (draft)
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	